

Roll No.

TME-701

**B. TECH. (MECH.) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, 2022**

MECHANICAL VIBRATIONS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the terms :

(CO1)

(i) Degree of Freedom

(ii) Resonance

(iii) Damping Ratio

(iv) Natural Frequency

(v) Time Period

(b) Add the following harmonic motions analytically and check the solution graphically :

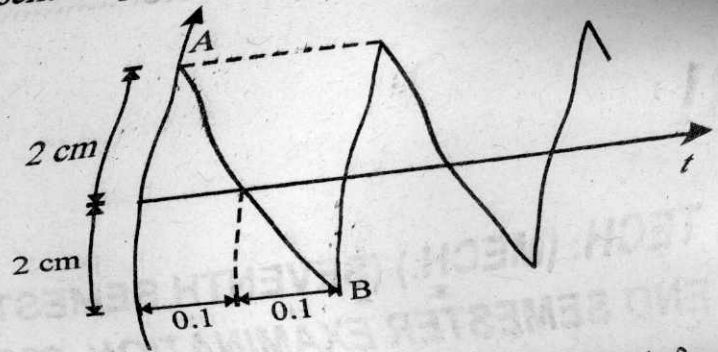
(CO1)

$$x_1 = 8 \sin(\omega t + 30^\circ)$$

$$\text{and } x_2 = 10 \sin(\omega t - 60^\circ)$$

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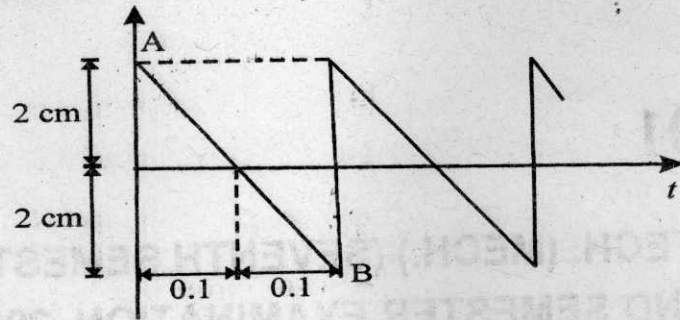
(c) Represent the periodic motion shown in figure by Harmonic series



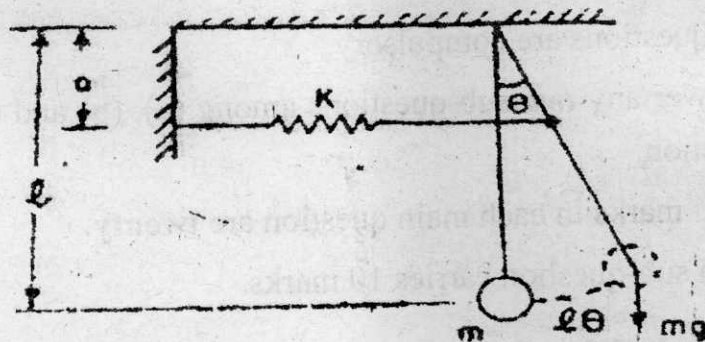
2. (a) Determine the equations of the motion and natural frequency of the simple pendulum mass m .

(c) Represent the periodic motion shown in figure by Harmonic series :

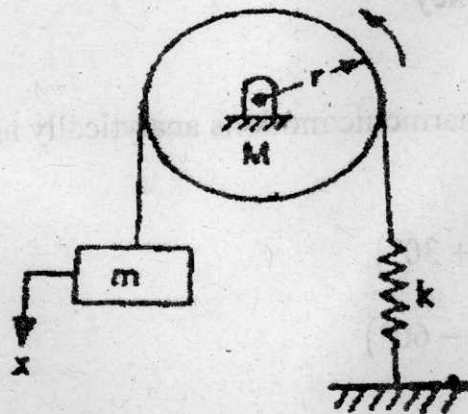
(CO1)



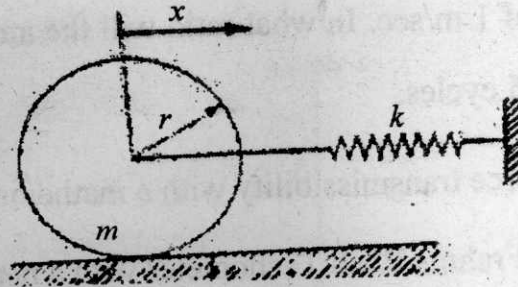
2. (a) Determine the equations of the motion and natural frequency of the simple pendulum mass m , length l , spring of stiffness k attached at distance a , from hanging point. (CO2)



- (b) Determine the equations of the motions and natural frequency of the pulley with mass M , mass moment of inertia I , radius r , attached with mass m at radius r and spring of stiffness k at radius r . (CO2)



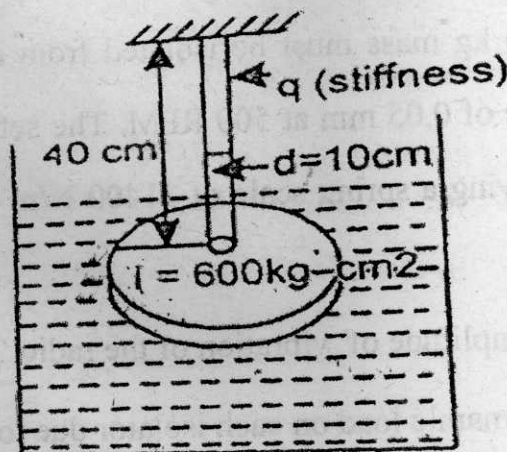
- (c) Determine the equations of the motions and natural frequency of the system shown in figure. (CO2)



3. (a) (i) Define Critical damping with a mathematical expression.
 (ii) Define Logarithmic decrement with a mathematical expression.
 (iii) Define Eddy current damping. (CO3)
- (b) Figure shows a torsional pendulum with a disc of moment of inertia $I = 600 \text{ kg cm}^2$. The pendulum is immersed in a viscous liquid. During vibrations of pendulum, the successive amplitude decays with 66% of initial value. Take $G = 4.4 \times 10^{10} \text{ N/m}^2$ for shaft. (CO3)

Determine :

- (i) logarithmic decrement
 (ii) damping torque per unit velocity
 (iii) the periodic time of vibration
 (iv) The frequency when the disc is removed from this liquid.



- (c) A body of mass 5 kg is supported on a spring of stiffness 1960 N/m and has dashpot connected to it which produces a damping force of 1.96 N at a velocity of 1 m/sec. In what ratio will the amplitude of vibration be reduced after 5 cycles. (CO3)

4. (a) (i) Define Force transmissibility with a mathematical expression.

(ii) Describe Frahm's Reed Tachometer with a neat sketch. (CO4, CO6)

- (b) A system having rotating unbalance has total mass of 25 kg. The unbalanced mass of 1 kg rotates with a radius 0.04 m. It has been observed that at a speed of 1000 rpm, the system and eccentric mass have a phase difference of 90° and the corresponding amplitude is 0.015 m. Find out : (CO4, CO6)

(i) natural frequency of the system

(ii) damping factor

(iii) amplitude at 1500 rpm

(iv) phase angle at 1500 rpm

- (c) A radio set of 20 kg mass must be isolated from a machine vibrating with an amplitude of 0.05 mm at 500 RPM. The set is mounted on four isolators, each having a spring scale of 31400 N/m and damping factor of 392 N-sec/m. (CO4, CO6)

(i) What is the amplitude of vibration of the radio ?

(ii) What is the dynamic load on each isolator due to vibrations ?

(5)

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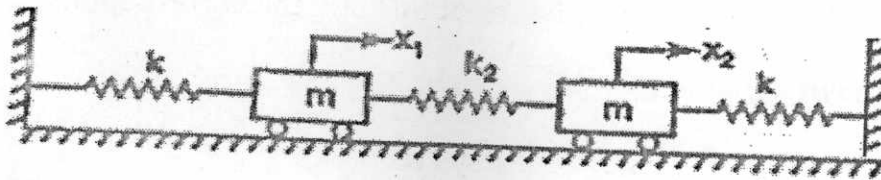
5. (a) Find the frequencies and mode shapes of the system shown in Fig. (Double pendulum) :

(CO5)



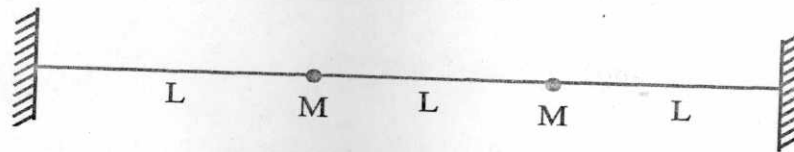
- (b) Find the frequencies and mode shapes of the system shown in Fig.

(CO5)



- (c) Find the frequencies of mode shapes of the system shown in Fig. (two masses fixed on a tightly stretched string).

(CO5)



Roll No.

TME-704

B. TECH. (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

OPERATION RESEARCH AND OPTIMIZATION TECHNIQUES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

1. (a) Discuss in brief, the role of OR models in decision-making. What areas of OR have made a significant impact on the decision-making process ?

(CO1)

(b) Solve the following LP problems using the simplex method :

(CO1)

Maximize :

$$Z = x_1 + x_2 + x_3$$

STC :

$$4x_1 + 5x_2 + 3x_3 \leq 15$$

$$10x_1 + 7x_2 + x_3 \leq 2$$

$$x_1, x_2, x_3 \geq 0.$$

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(c) Use penalty (Big-M) method to solve the following LP problem : (CO1)

Min. :

$$Z = 5x_1 + 2x_2 + 10x_3$$

STC :

$$x_1 - x_3 \leq 10$$

$$x_2 + x_3 \geq 10$$

$$x_1, x_2, x_3 \geq 10.$$

2. (a) A departmental store wishes to purchase the following quantities of sarees : (CO2)

Types of sarees	Quantity
A	150
B	100
C	75
D	250
E	200

Tenders are submitted by four different manufacturers who undertake to supply not more than the quantities mentioned below (all types of sarees combined) :

Manufacturer	Total Quantity
W	300
X	250
Y	150
Z	200

		Sarees				
		A	B	C	D	E
Manufacturer	W	275	350	425	225	150
	X	300	325	450	175	100
	Y	250	350	475	200	125
	Z	325	275	400	250	175

How should the orders be placed ?

- (b) A project work consists of four major jobs for which an equal number of contractors have submitted tenders. The tender amount quoted (in lakh of rupees) is given in the matrix. (CO2)

		Job			
		a	b	c	d
Contractor	1	10	24	30	15
	2	16	22	28	12
	3	12	20	32	10
	4	9	26	34	16

Find the assignment which minimizes the total cost of the project when each contractor has to be assigned at least one job.

- (c) A head of department in a college has the problem of assigning courses to teachers with a view to maximize educational quality in his department. He has available to him one professor, two associated professors and one teaching assistant (TA). Four courses must be

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offered. After appropriate evaluation, he has arrived at the following relative rating (100 = best rating) regarding the ability of each instructor to teach each of the four courses : (CO2)

	Course 1	Course 2	Course 3	Course 4
Prof. 1	60	40	60	70
Prof. 2	20	60	50	70
Prof. 3	20	30	40	60
TA	30	10	20	40

How should he assign his staff to the courses in order to realize his objective ?

3. (a) A research and development department is developing a new power supply for a console television set. It has broken the job down into the following : (CO3)

Job	Description	Immediate Predecessors	Time (days)
A	Determine output voltages	—	5
B	Determine whether to use solid state rectifiers	A	7
C	Choose rectifier	B	2
D	Choose filters	B	3
E	Choose transformer	C	1
F	Choose chassis	D	2
G	Choose rectifier mounting	C	1
H	Layout chassis	E, F,	3
I	Build and test	G, H	10

- (i) Draw the network diagram of activities involved in the project and indicate the critical path.
- (ii) What is the minimum completion time for the project ?
- (b) The owner of a chain of fast restaurants is considering a new computer system for accounting and inventory control. A computer company sent the following information about the computer system installation.

(CO3)

Activity	Activity Description	Immediate Predecessor	Times (days)		
			Optimistic	Most Likely	Pessimistic
A	Select the computer model	—	4	6	8
B	Design input/output system	A	5	7	15
C	Design monitoring systems	A	4	8	12
D	Assemble computer hardware	B	15	20	25
E	Develop the main programmes	B	10	18	26
F	Develop input/output routines	C	8	9	16
G	Create database	E	4	8	12
H	Install the system	D, F	1	2	3
I	Test and implement	G, H	6	7	8

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- (c) Find an optimal sequence for the following sequencing problem of four jobs and five machines (when passing is not allowed) of which processing time (in hrs) is as follows : (CO5)

Job	1	2	3	4
Machine M_1	6	5	4	7
Machine M_2	4	5	3	2
Machine M_3	1	3	4	2
Machine M_4	2	4	5	1
Machine M_5	8	9	7	5

Also find the total elapsed time.

5. (a) Two jobs are to be processed on four machines A, B, C and D. The technological order for these jobs on machines is as follows : (CO5)

Job 1	Job 2
A	D
B	B
C	A
D	C

Processing times are given in the following table :

		Machines			
		A	B	C	D
Job 1		4	6	7	3
Job 2		4	7	5	8

Find the optimal sequence of jobs on each of the machines.

- (b) Derive the difference equations for the queuing model $\{(M/M/1) : (\infty/FCFS)\}$. How would you proceed to solve the model? (CO5)
- (c) Consider a single server queuing system with Poisson input and exponential service times. Suppose the mean arrival rate is 3 calling units per hour, the expected service time is 0.25 hour and the maximum permissible calling units in the system is two. Derive the steady-state probability distribution of the number of calling units in the system, and then calculate the expected number in the system. (CO6)

Roll No.

TME-706

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

POWER PLANT ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each question carries 10 marks.

1. (a) What is doubling time in power plant engineering ? How is the load-duration curve constructed ? (CO1)

(b) What do you understand by depreciation ? How is the depreciation rate determined ? (CO1)

(c) A power plant has the following annual factors : (CO1)

Load factor = 75%, Capacity factor = 60%, Use factor = 65%,
Maximum demand = 60 MW.

Estimate :

(i) Annual energy production

(ii) Reserve capacity over and above the peak load

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2. (a) Give the layout of modern steam power plant. Explain all the circuits. (CO2)
(b) Discuss various methods of compounding in steam turbines. (CO2)
(c) Steam with absolute velocity of 300 m/s is supplied through a nozzle to a single stage impulse turbine. The nozzle angle is 25° . The mean diameter of the blade rotor is 1 meter and it has a speed of 2000 r.p.m. Find suitable blade angles for zero axial thrust. If the blade velocity coefficient is 0.9 and the steam flow rate is 10 kg/s, calculate the power developed. (CO2)
3. (a) Explain the essential components of a Diesel Power Plant. (CO3, CO4)
(b) Derive the optimum pressure ratio in an ideal gas turbine for maximum work output. (CO3, CO4)
(c) Draw the layout of Gas Power Plant. List the advantages and limitations of gas power plant. (CO3, CO4)
4. (a) State the essential elements of Hydroelectric Power Plant ? Explain the different methods of classifying a Hydroelectric Power Plant. (CO5)
(b) Explain with neat sketch the principle of operation of Pelton Turbine. (CO5)
(c) A pelton wheel has a mean bucket speed of 10 m/s with a jet of water flowing at a rate of $0.7 \text{ m}^3/\text{s}$ under a head of 30 m. The buckets deflect the jet through an angle of 160° . Calculate the power and efficiency of the turbine. Take $C_v = 0.98$. (CO5)
5. (a) Write short notes on Boiling Water Reactor (BWR) and Pressurised Water Reactor (PWR). (CO6)
(b) Calculate the mass defect and binding energy per nucleon (MeV) of oxygen. Given : $m_p = 1.007277 \text{ amu}$, $m_n = 1.0086665 \text{ amu}$, $m_e = 0.00055 \text{ amu}$, atomic mass of oxygen = 15.99491 amu. (CO6)
(c) What is CANDU-type reactor ? Explain with neat diagram. (CO6)

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TME-707

**B. TECH. (ME) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

AUTOMOBILE ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Explain in detail about various types of resistances offered to the vehicle while in motion. (CO1)
- (b) Discuss in detail about various components of Internal Combustion engine along with their functions with the help of neat labelled diagram. (CO1)
- (c) Discuss in detail about the need of a 'Gearbox' in a vehicle. (CO1)
2. (a) Discuss in detail about various components of Electronic Fuel Injection (EFI) system. (CO2)

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- (b) Explain in detail about engine emission control using three way catalytic converter systems. (CO2)
- (c) Differentiate between Common rail direct injection system and Rotary distributor type fuel injection system. (CO2)
- 3. (a) Discuss in detail about 'Sliding mesh gearbox' and 'Constant mesh gearbox' system with the help of neat labelled diagram. (CO3)
- (b) Explain in detail about construction and working of 'Hotchkiss Drive' and 'Torque Tube Drive'. (CO3)
- (c) Discuss in detail about construction and working of 'Torque converter' and 'Propeller shaft'. (CO3)
- 4. (a) Discuss in detail about Steering geometry of an automobile. (CO4, CO5)
- (b) Discuss in detail about construction and working of hydraulic and electric power steering systems. (CO4, CO5)
- (c) Differentiate between 'independent suspension' system and 'leaf spring suspension' systems. (CO4, CO5)
- 5. (a) Explain in detail about construction, working and benefits of an 'Electric vehicle'. (CO6)
- (b) Discuss in detail about usage, engine modification, and benefits of using 'natural gas' as an alternative fuel in an automobile. (CO6)
- (c) Discuss in detail about usage, engine modification, and benefits of using 'bio-ethanol' as an alternative fuel in an automobile. (CO6)

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TME-714

**B. TECH. (ME) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

TOTAL QUALITY MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Explain eight dimensions of product quality. (CO1, CO4, CO2)
(b) Differentiate between accuracy and precision with *five* points.
(CO1, CO4, CO2)
(c) Explain *ten* barriers to TQM. (CO1, CO4, CO2)
2. (a) What are quality statements ? Explain with *two* examples.
(CO1, CO2, CO5, CO6)
(b) What is cause and effect diagram ? Explain in detail with example.
(CO1, CO2, CO5, CO6)

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- (c) You want to select a supplier of tires for your company which is in manufacturing of tractors. Explain the selection criteria (5 point) for selection of the same. (CO1, CO2, CO5, CO6)
3. (a) What is performance appraisal ? What you will consider for performance appraisal of the team under you involved in design of lathe machines (5 points) ? (CO6, CO2, CO3)
- (b) Explain poka-yoke with any *two* examples. (CO6, CO2, CO3)
- (c) Why do we need benchmarking ? Explain with advantages (5 points). (CO6, CO2, CO3)
4. (a) Compare kaizen and innovation with *five* points. (CO4, CO2)
- (b) What is quality circle ? Explain the advantages. (CO4, CO2)
- (c) Explain the cost of quality in detail. (CO4, CO2)
5. (a) Explain quality function deployment (QFD) with diagram. (CO1, CO2, CO3)
- (b) Explain ISO 9001-2000 in detail. (CO1, CO2, CO3)
- (c) What are the types of quality audits ? Explain. (CO1, CO2, CO3)

Roll No.

DETF-03

B. TECH. (ME) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2022

REFRIGERATION AND AIR CONDITIONING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the following terms : (CO1)

- (i) Refrigeration,
- (ii) Air conditioning,
- (iii) 1 TR, and
- (iv) Significance of COP.

(b) A simple air cooled system is used for an aeroplane having a load of tonnes. The atmospheric pressure and temperature are 0.9 bar and 10°C respectively. The pressure increases to 1.013 bar due to ramming. The temperature of the air is reduced by 50°C in the heat exchanger. The pressure in the cabin is 1.01 bar and the temperature of air leaving cabin is 25°C. (CO3, CO6)

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Determine :

- (i) Power required to take the load of cooling in the cabin; and
- (ii) C.O.P. of the system.

Assume that all the expansions and compressions are isentropic. The pressure of the compressed air is 3.5 bar.

- (c) Write the advantages and disadvantages of air refrigeration system. Also, explain the simple air cooling refrigeration cycle with the help of a neat sketch. (CO2)
2. (a) refrigerator operates between temperature limits of 30°C and -5°C . The refrigerant is 0.97 dry before leaving the evaporator coil. Find the condition of refrigerant entering the evaporator and COP of system.

(CO3, CO6)

Temp.	Enthalpy, kJ/kg		Entropy, kJ/kg.K		Specific heat, kJ/kg.K	
$^{\circ}\text{C}$	h_f	h_g	s_f	s_g	$C_{p,L}$	$C_{p,g}$
30	323.22	1465.38	1.2037	4.9839	5.024	3.35
-5	158.26	1431.89	0.63	5.4072	—	—

Take C_p for superheated vapour as 3.35 kJ/kg.K .

- (b) Draw a two stage compression system with liquid intercooler and explain its working with the help of a $p-h$ diagram. (CO2, CO3)
- (c) Explain various factors affecting the performance of VCRC.

(CO2, CO3)

(3)

3. (a) How are the refrigerants classified ? Explain the each type with example. Also, write the name of substitutes for CFC refrigerants and their importance. (CO1)
- (b) Designate the following refrigerants : CHCl_2CF_3 , C_2H_4 and CO_2 . Also, explain the thermodynamic properties of refrigerants. (CO1, CO2)
- (c) Draw a neat sketch of a three fluid (Electrolux) vapour absorption refrigeration system and explain its working. (CO2, CO4)
4. (a) Write the differences between vapour compressor refrigeration system and vapour absorption refrigeration system. Also explain the simple ammonia-water absorption refrigeration system with the help of a neat diagram. (CO2, CO4)
- (b) Classify the refrigerant compressors and explain each type in brief. (CO2)
- (c) What do you mean by Cryogenics ? Explain Cascade refrigeration systems with a neat diagram. (CO2, CO4)
5. (a) What do you mean by psychrometry ? Explain various psychrometric processes with the help of a psychrometry chart. (CO5)
- (b) Atmospheric air at 95 kPa, 30°C has a relative humidity of 70%. Determine humidity ratio. The saturation pressure of water at 30°C is 4.246 kPa. (CO5, CO6)
- (c) Explain the working of a summer air conditioner with the help of a diagram. (CO4, CO5)

Roll No.

TAS-708

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

FLIGHT DYNAMICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Define coefficient of lift. How does it behave after the airfoil has stalled ? (CO1)
- (b) Will it be easier to an airplane in troposphere or stratosphere, why ? (CO1)
- (c) Why do you think a turbofan has better efficiency than a turbojet ? (CO1)
2. (a) Draw a diagram and show the forces acting on a glider. What is glide ratio and why it is a ratio of coefficient of lift to coefficient of drag ? (CO1, CO2)

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- (b) Explain the absolute and service ceiling for an aircraft. (CO1, CO2)
- (c) Derive the expression for range of an aircraft. (CO1, CO2)
- 3. (a) Explain with diagram what you understand by static and dynamic stability. (CO2, CO3)
- (b) Draw the graph between the pitching moment and angle of attack for a stable fixed wing aircraft. Also explain it. (CO2, CO3)
- (c) What are canards ? Draw necessary diagram. What is their advantage and disadvantage ? (CO2, CO3)
- 4. (a) Explain phugoid mode in detail. (CO3, CO4)
- (b) With a diagram derive the equation for the contribution of moment to the CG of an aircraft due to the wing. (CO3, CO4)
- (c) Explain why a swept back wing will improve the directional stability of an airplane. (CO3, CO4)
- 5. (a) Explain elliptic, ballistic and continuous thrust ascent with diagram. (CO5, CO6)
- (b) Explain Hohmann transfer in short. (CO5, CO6)
- (c) Draw diagram of an elliptical orbit and give expression for the velocity at apogee and perigee. (CO5, CO6)

Roll No.

TAE-707

**B. TECH. (ME) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
AUTOMOTIVE CHASSIS AND SUSPENSION**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is a self-locking differential ? What are its types ? (CO1)
(b) Explain the difference between body centre of gravity and vehicle centre of gravity with the help of a simple sketch. (CO1)
(c) Write notes on the following : (CO1)
 - (i) Universal joint
 - (ii) Constant speed joint
2. (a) What is a McPherson strut suspension ? What are its advantages ? (CO2)
(b) What are the advantages and disadvantages of a front-wheel drive ? (CO2)
(c) What are the functions of shock absorber/suspension dampers ? Differentiate between twin-tube and mono-tube dampers. (CO2)

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3. (a) With the help of neat sketches, explain the construction and application of : (CO3)
- (i) Diagonal ply tyres
 - (ii) Radial ply tyres.
- (b) A tyre is marked as [185/60 R14 80S] to be mounted on wheel 6J × 14. Explain what each number and character represent in this marking. (CO3)
- (c) Explain the following terms with suitable sketches : (CO3)
- (i) Camber
 - (ii) Toe-in
 - (iii) Roll Axis
4. (a) Explain the construction and working of a Rack and Pinion steering system. What are its advantages and disadvantages ? (CO4)
- (b) What is meant by a Power steering system ? Explain the working of an Electrical Power steering system. (CO4)
- (c) What is meant by steering column ? Explain any *two* steering tube configurations designed to meet the safety requirements of a passenger car. (CO4)
5. (a) What are the functions of a braking system ? Explain the design and construction of a Fixed caliper disc brake. (CO5)
- (b) Explain the working of a hydraulic braking system. What are the conditions to be satisfied by a braking fluid ? (CO5)
- (c) From the point of view of vehicle structural safety, explain the working of : (CO5)
- (i) Air bags
 - (ii) Seat belts
 - (iii) Passenger compartment integrity.

Roll No.

TRA-707

**B. TECH. (ME-RA) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

AUTOMATIC CONTROL

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

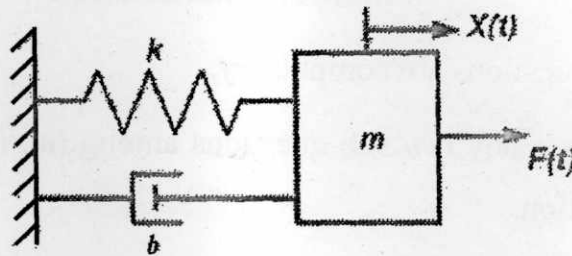
(iv) Each question carries 10 marks.

1. (a) Describe the open-loop and closed-loop systems with neat block diagrams. Also, discuss their advantages and disadvantages. (CO1)
- (b) Plot and write the expression (with Laplace transform) for the following signals : step, ramp, sinusoidal, and impulse. (CO1)
- (c) Derive the inverse Laplace of the following function : (CO1)

$$F(s) = \frac{2s + 12}{s^2 + 2s + 5}$$

P. T. O.

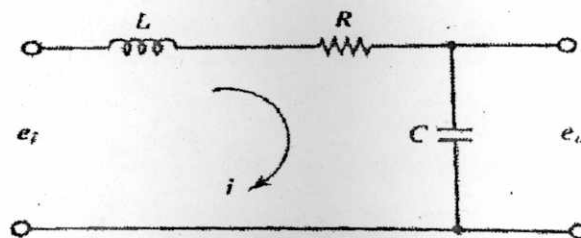
2. (a) What is the characteristic equation of a system ? Describe the following system responses : undamped, overdamped, underdamped, and critically damped systems in terms of roots of the characteristic equation. (CO2 & CO3)
- (b) Linearize the following nonlinear equation around an operating point $x = 2$: $y = 4x^3 + 5$. Also, check the error, if the operating point is changed to $x = 3$. (CO2 & CO3)
- (c) Derive the transfer function of the mass-spring dashpot system as shown in the figure below. Also, comment about the stability and behaviour of the system if, $m = 1$, $b = 3$, and $k = 2$. (CO2 & CO3)



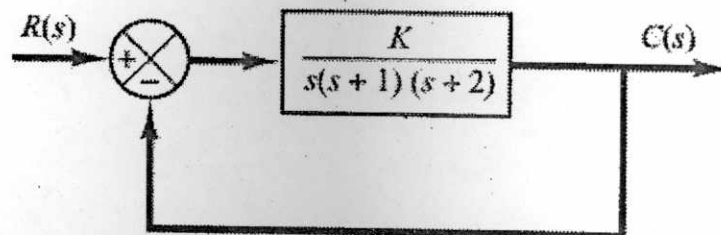
3. (a) Derive the expression for the unit-step response of a first-order system. (CO2 & CO3)
- (b) Comment about the stability of a system using Routh's stability criterion, whose characteristics equation is given by the following polynomial : (CO2 & CO3)
- $$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0.$$
- (c) Describe the following transient-response characteristics of a control system with a neat sketch : (CO2 & CO3)
- (i) Delay time

- (ii) Rise time
- (iii) Peak time
- (iv) Maximum overshoot
- (v) Settling time

4. (a) What are the advantages of the modern control theory compared to the classical control theory ? Also, describe the state space representation of a dynamic system. (CO4)
- (b) Derive the state-space model of the R-L-C circuit shown in the figure below. Consider input is e_i and output is e_o . (CO4)

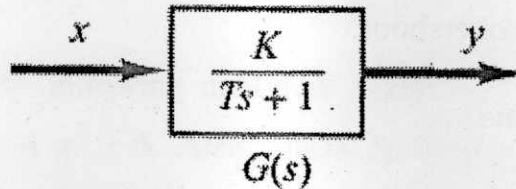


- (c) Plot the root locus of the following closed-loop system : (CO4)



P. T. O.

5. (a) Obtain the steady state output y of the system shown in the figure below for a given sinusoidal input $x = A \sin(\omega t)$. (CO5 & CO6)



- (b) Draw Bode plots for the integral term $G(s) = K/s$. (CO5 & CO6)
- (c) Describe the Proportional-Integral-Derivative (PID) control action with governing equations. (CO5 & CO6)

Roll No.

TAE-714

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
VEHICLE BODY ENGINEERING AND SAFETY**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How can aluminium alloys improve vehicle safety and fuel efficiency ?
Justify your answer with suitable data. (CO1, CO3, CO4)
- (b) Define the importance of the extrusion process in automotive parts manufacturing. List a few automotive components made using the extrusion process. (CO1, CO3, CO4)
- (c) What makes composites so popular in automotive applications ?
Classify various automotive composites. (CO1, CO3, CO4)
2. (a) "Ergonomics Is an Important factor in vehicle design." Critically examine this statement. (CO2, CO4, CO5)

P. T. O.

(2)

- (b) Why is good vehicle packaging important for an automobile ? Give an appropriate example. (CO2, CO4, CO5)
- (c) Show the "blind spots" of a sedan. How will you conduct the driver's visibility test ? (CO2, CO4, CO5)
3. (a) What are the various aerodynamic forces that act on an automobile at high vehicle speeds ? Explain any *one* in detail. (CO3, CO5, CO6)
- (b) Illustrate the yawing, pitching, and rolling of a four-wheeler along with a brief explanation of each. (CO3, CO5, CO6)
- (c) Draw a wind tunnel and explain the purpose of a wind tunnel in vehicle design. (CO3, CO5, CO6)
4. (a) What are the various techniques through which we can reduce the various vehicle noises ? (CO1, CO5, CO2)
- (b) How will you improve an automobile's impact safety ? Explain all the possible techniques. (CO1, CO5, CO2)
- (c) Explain the effect of mass distribution and engine location on stability in detail. (CO1, CO5, CO2)
5. (a) Classify various types of automobiles along with their applications. (CO1, CO2, CO5, CO6)
- (b) What are the various safety features of an automobile ? Why are vehicle safety considerations important ? (CO1, CO2, CO5, CO6)
- (c) What are the design criteria based on passenger capacity and goods to be transported over long distances ? (CO1, CO2, CO5, CO6)

Roll No.

TRA-714

B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
SENSORS APPLICATION IN MANUFACTURING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the role of sensors in any system and define some basic terminologies. (CO1)
- (b) Differentiate between the sensors and transducers. (CO1)
- (c) Discuss the working principle of sensor and transducer with static and dynamic characteristics. (CO1)
2. (a) What are operational amplifiers, filters, and signal converters ? (CO2)
- (b) Discuss about the following : (CO2)
 - (i) Inductive sensor
 - (ii) Photo sensor
 - (c) Magnetic sensors

P. T. O.